

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B.Tech./MBA Tech/BTI

Year: III Semester: VI

Stream/s: All applicable streams

Subject: Elements of Project Management

No. of Pages 08

SYNOPTIC SOLUTION

Year III Semester IV

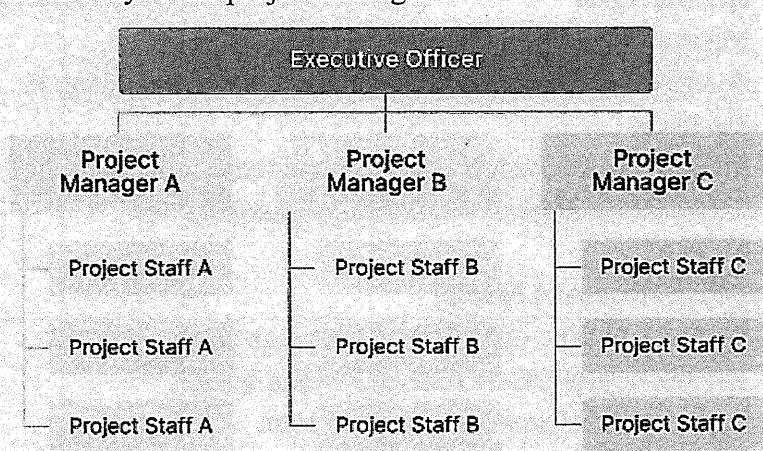
Programme: B.Tech. MBA (Tech Mgt)

Course: All applicable courses

Subject: Business of Project Management

No. of Pages: 28

SYNOPSIS SOLUTION

Q1		Marks
a.	Project may be defined as “ A temporary endeavor undertaken to create a unique product, service, or result” OR Project is defined as temporary but interrelated tasks undertaken to give a unique product or service or result List of Four Projects [Definition:02 Marks, List of completed mega project: 2Marks]	4
b.	-Projectized organization is a project focused organizational structure where project manager has the final authority over the project to make project decisions, priorities, acquire and assign resources -A projectized organization refers specifically to an organizational structure that has been set up in a manner in which the project manager leads the group and in which the project manager has the ultimate authority to make any and all decisions involving the organization -In a projectized structure, all the work is looked at as a project, the project manager has complete control, all team members report directly to the project manager  <pre> graph TD EO[Executive Officer] --> PM_A[Project Manager A] EO --> PM_B[Project Manager B] EO --> PM_C[Project Manager C] PM_A --> PS_A1[Project Staff A] PM_A --> PS_A2[Project Staff A] PM_A --> PS_A3[Project Staff A] PM_B --> PS_B1[Project Staff B] PM_B --> PS_B2[Project Staff B] PM_B --> PS_B3[Project Staff B] PM_C --> PS_C1[Project Staff C] PM_C --> PS_C2[Project Staff C] PM_C --> PS_C3[Project Staff C] </pre>	4
c.	List of purposes of project planning- 1. To clearly identify and describe business requirements 2. To establish cost, schedule, list of deliverables and delivery dates 3. Resources planning for their effective utilization 4. Obtaining top management approval and proceeding to the next phase	4
d.	-Positive risks are risks that result in good things happening; sometimes called opportunities [01Marks] -Negative risk involves understanding potential problems that	4

	might occur in the project and how they might impede project success [01Marks] Technical Risks - 1.refers to the failure to meet a particular performance requirement 2.failure of the feasibility of a design and changes in technology are some of the sources of technical risks 3. the project is under technical risk if the technology being used becomes outdated by the time the project delivers the final product/service (particularly in software projects), this forces project manager to continuously update the technical aspects of project [02Marks]	
e	List of reasons- 1. End the external contracts of the company formally with the vendors, clients and other parties 2. Suspend the assignments of the team member formally 3. Seek the acceptance of the client formally on the project work and the output 4. Make sure of proper installation and/or implementation of the delivered output on time within budget and as per specifications 5. Make sure of the availability of sufficient project documentation that can facilitate any change occurring in the future 6. Submit the final report and get it approved 7. Close the established relationships internally and externally [4 to 5 reasons: 4 Marks]	4
Q2a	1. To control scope of project and manage change 2. To deliver projects on time and within budget 3. To ensure the focus of the project team 4. To collect user requirement from disparate sources 5. To define the critical path to optimally complete the project 6. To provide a process for estimating project resources, time and costs 7. To communicate project progress, risks and changes 8. To explore project assumptions 9. To prepare for unexpected project issues 10. To document the knowledge gained from the project [6 to 7 points along with explanation = 10Marks]	10

Q2b	List of parameters used for screening candidates- 1. Background and experience 2. Leadership and strategic expertise 3. Technical expertise 4. People management skill -Selection is difficult as both formal, informal and personal parameters/skills/qualifications need to be evaluated - Some desired personal characteristics for Project Manager need to be checked (and also formal qualifications/skill sets) 1. Flexibility and adaptability 2. Preference for significant initiative and leadership 3. Aggressiveness, confidence, persuasiveness, verbal fluency 4. Ambition, activity, forcefulness 5. Effectiveness as a communicator and integrator 6. Broad scope of personal interests 7. Poise, enthusiasm, imagination, spontaneity 8. Able to balance technical solutions with time, cost, and human factors 9. Well organized and disciplined 10. A generalist rather than a specialist 11. Able and willing to devote most of his time to planning and controlling 12. Able to identify problems 13. Willing to make decisions 14. Able to maintain proper balance in the use of time 15. Emotional intelligence (Along with formal skill sets at least 4 to 6 points for personal characteristics) [List:03 Marks, Explain: 07Marks]	10

Q3a	1. What? The process defines the project <i>objectives</i>, <i>scope</i>, and <i>system requirements</i>. These specify the project deliverables, end-items, and other sought results, as well as the time, cost, and performance targets. 2. How? It defines the <i>work activities</i>, tasks, or jobs to be done to achieve the objectives and requirements. These activities include everything necessary to create and deliver the end-item or deliverables, including planning, control, and administration activities. 3. Who? It specifies the <i>project organization</i>—the individuals or departments, subcontractors, and managers that will perform and manage the work, and specifies their responsibilities. 4. When, in what order? It creates a <i>schedule</i> showing the timing of work activities, deadlines, and milestone dates. 5. How much and when? The process creates a <i>budget</i> and <i>resource plan</i> to fund and support the project. 6. How well? It specifies a method for tracking and controlling project work, which is necessary to keep the project conforming to the schedule, budget, and user and system requirements.	10
Q3b	-Project Risk is an uncertain event or condition that, if it occurs has negative or positive effect on one or more project objectives -Project risk management process helps in identifying, evaluating and planning responses to events/risks, both negative and positive that might occur throughout the project life cycle -Project risk management helps to reduce/eliminate negative risks and its consequences and make the things in the best interests of meeting project objectives -Risk tolerance or risk utility is the amount of satisfaction or pleasure received from a potential payoff Risk-Averse Risk-Neutral Risk-Seeking utility potential payoff utility potential payoff utility potential payoff -Utility rises at a decreasing rate for people who are risk-averse -Those who are risk-seeking have a higher tolerance for risk and their satisfaction increases when more payoff is at stake -The risk-neutral approach achieves a balance between risk and payoff [Need: 3 Marks ,Detail explanation with diagram diagram:07Marks]	10

Q4a	1. Sketch the network diagram – 03 Marks 2. critical path and total duration of the project -02Marks 3. Earliest start time, Earliest finish time, Latest start time and Latest finish time for each activity – 04 Marks 4. Evaluate total float for all activities – 03 Marks	12
Q4b	-SWOT(Strengths Weaknesses, Opportunities, and Threats) analysis, also known as the SWOT matrix, is one of the techniques project managers and organizations have implemented in their risk management practices to identify the risk. - SWOT analysis framework provides a structured approach to identifying and assessing risks. <i>Utilising strengths to identify risks-</i> -While strengths are typically seen as positive attributes, they can also present risks. For example, if an organisation heavily relies on a single supplier, the sudden disruption in their operations can pose a significant risk -By analysing strengths critically, organisations can identify potential vulnerabilities and mitigate associated risks <i>Identifying risks through weaknesses-</i> -Weaknesses inherently expose organisations to risks -By recognising and describing weaknesses, organisations gain insights into areas where they are most susceptible -For instance, if a project lacks adequate resources, it may face risks related to delays or quality issues -By addressing weaknesses, organisations can proactively manage and mitigate risks <i>Exploring risks within opportunities-</i> -Opportunities may carry inherent risks that need to be identified and described -For instance, entering a new market presents growth opportunities but also introduces risks related to competition, regulatory compliance, or technological challenges -By assessing these risks within opportunities, organisations can develop strategies to address them effectively <i>Identifying threats through threat component-</i> -Most people will focus on threats when identifying their risks -A threat is something that could go wrong so it is normal for organisations to focus on avoiding or reducing negative outcomes when forward planning -The threats component of the SWOT analysis allows organisations to identify internal risks -By examining threats, such as new market entrants or changes in	8

	regulations, organisations can anticipate potential risks and take proactive measures to mitigate their impact	
Q5a	-It is necessary to Track or Measure the progress we have achieved towards a Goal we wish to accomplish -Industries uses Project Management to Aid in Maximizing and Optimizing resources to accomplish set goals and objectives - Explanation of National Covid-19 Vaccination Programme- India with facts, figures and project management perspectives [Need:02 Marks, Explanation for Covid-19 project:08Marks]	10
Q5b	-PMBOK defines Work Breakdown Structures as a deliverable – oriented grouping of activities that organizes and defines the total scope of project [02Marks] <i>Benefits of WBS-</i> 1. Tasks are definable or easily understood 2. Tasks becomes independent 3. Time & cost can be estimated easily 4. Progress becomes measurable 5. Tasks can be adaptable 6. Simplification in risk identification 7. Improved accountability 8. Enhanced visibility [04Marks] Top level - project title Second level – control accounts Third level- work packages Fourth level – activities [02Marks] Example [02Marks]	10
Q6a	-Quantitative Risk Analysis is a project management process that numerically analyzes the effects of identified risks on the entire project objectives [02Marks] <i>Importance-</i> 1. Better overall project risk analysis 2. Better business decisions 3. Better estimates [03Marks] <i>Quantitative Risk Analysis is more often used for-</i> 1. Project that require a contingency reserve for the schedule and budget 2. Large,complex projects that require GO/NO-GO decisions which occurs multiples times in a project	8

	3. Projects where upper management wants more detail about the probability of completing the project on schedule and within budget [03 Marks]	
Q6b	<i>Steps-</i> 1. Defining the project scope precisely and clearly 2. Using relevant and reliable database 3. Designing an organization structure that is appropriate for the current project 4. Monitoring and controlling deviations from the project plan 5. Periodically evaluating and monitoring cost performances 6. Checking whether the changes are recorded in the cost baseline 7. Selecting vendors and project consultants carefully [Four steps: 04 Marks]	4
Q6c	-The cost baseline is defined as an output of cost budgeting a time phased budget that periodically measures and monitors the cost performance of budget OR - A cost baseline is the budget that has been approved for a project, broken down into a list of salaries, materials, equipment and more [02Marks] <i>What it includes-</i> 1. It includes every project activity, task and associated resources that are needed for a project 2. Any contingency reserve funds that are set aside to deal with potential issues that identified in a risk analysis [02Marks] <i>Importance-</i> 1. Setting a cost baseline is important because keeping to your project budget is important 2. This line helps to know whether you are going over budget which is a sure sign that your project is heading in the wrong direction 3. Helps to assess cost variance, cost performance and planned effort versus actual effort 4. Without a baseline, you have nothing to measure your spending against and control costs [04Marks]	8
Q7a	<i>Contents of project closing report-</i> 1. Business objectives 2. Closing statement 3. Measuring the benefits	8

	4. Risks, issues and outcomes 5. Efficiency of the project Explanation of above points- 08Marks	
Q7b	-The extra costs incurred over the estimated costs are called cost overruns [02 Marks] Impacts on project success- 1. Profitability: Cost overrun reduces the profitability of a project, as it implies that the project is spending more than it is earning 2. Quality: Cost overrun can compromise the quality of a project, as it may force the project team to cut corners, use cheaper materials, or reduce the scope or functionality of the project 3. Delivery: Cost overrun can delay the delivery of a project, as it may cause the project team to miss deadlines, extend schedules, or request more resources [Above three points with more explanation – 06 Marks]	8
Q7c	Relating different kind of projects and use of different project management concept/tools to manage project with successful outcomes	4